

Lunar Smart Adaptive Habitat Design with AI Psychological Monitoring for Crew Adaptive Positive Mental Health

2025-26

BIOMEDICAL
And / Or
Software Projects or Both

What we built

- Lunar Smart Adaptive Habitat Design with AI Psychological Monitoring
  - **Our design:** Two working Unity/VR simulators of our habitat, plus the physical scale model.



Realistic game



Sci-fi game

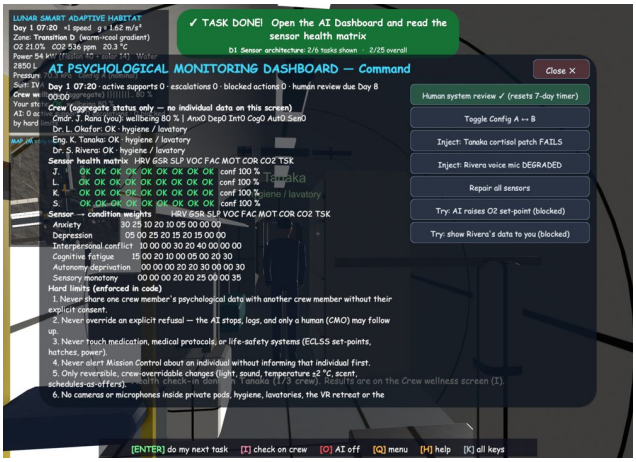


- Biomedical and / or software projects, or both
  - **Our design:** Both. Software: the AI engine and two games. Biomedical: simulated wearable data (heart rate, HRV, sleep, skin response, cortisol) for every crew member, which you collect with the health check-in.



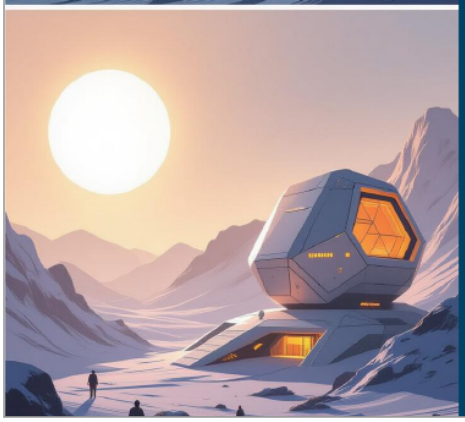
- Goal: psychological research + AI to monitor and support mental health
  - **Our design:** The AI scores six mental-health conditions live for each person and adapts the rooms to help.

- …while integrating smart building systems for life support and power
  - **Our design:** Life support (CO2 scrubber, water recycler, O2), solar + fission power and alarms are all in the same game. You even clean the CO2 scrubber yourself in a mission.



Realistic game: AI dashboard

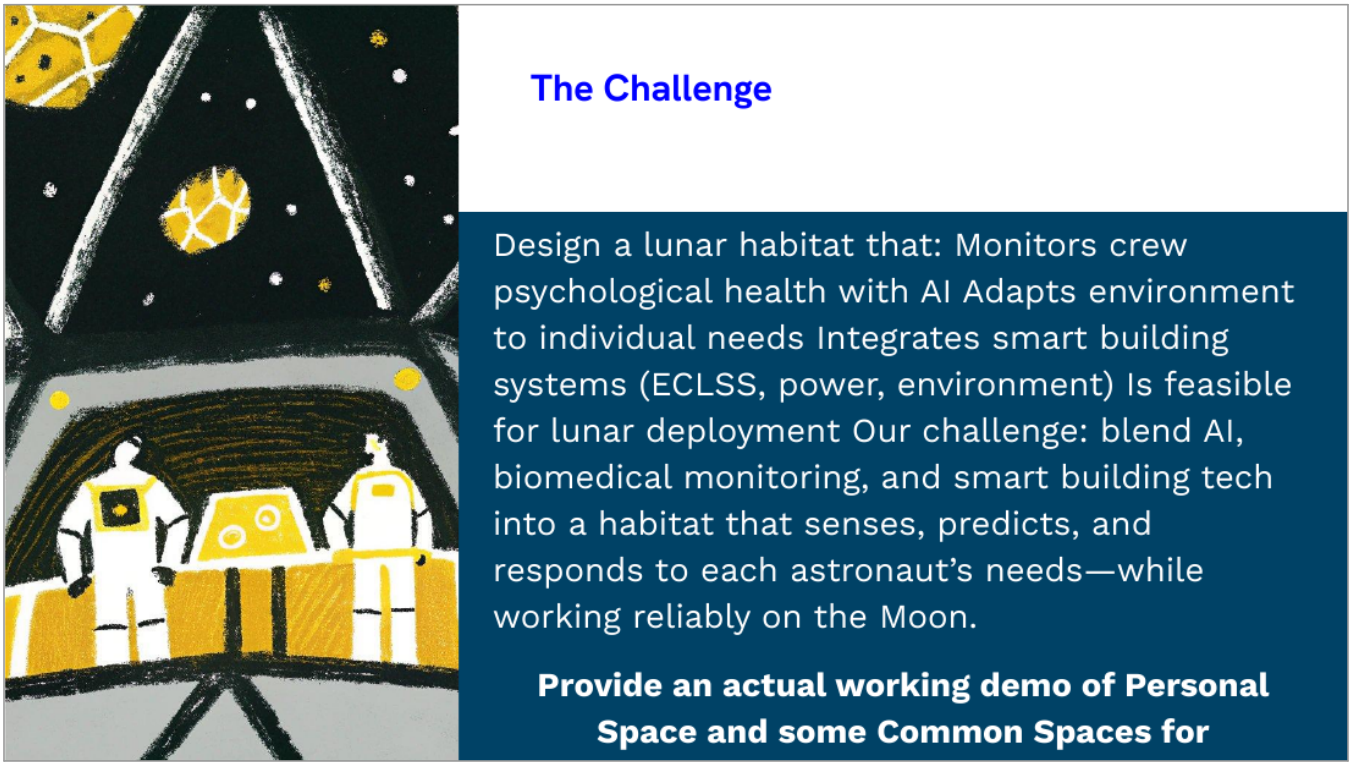




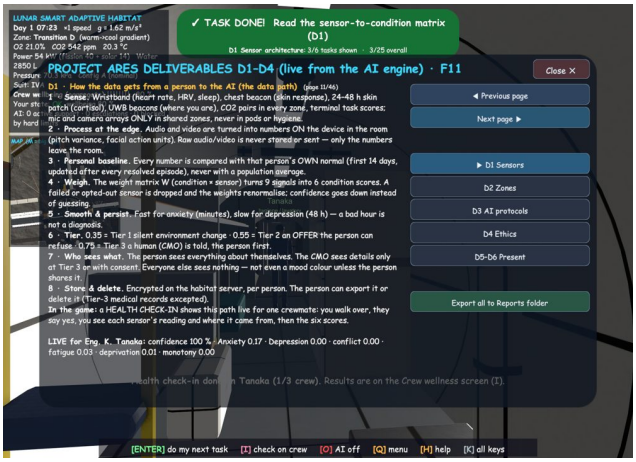
The Problem

Long lunar missions = isolation, stress, and risk to mental health Current habitats: focus on physical safety, not psychological needs Need: Adaptive, supportive environments for crew well-being Astronauts face unique psychological challenges: isolation, confinement, and stress. Traditional habitats protect physically, but we must also address mental health for mission success.

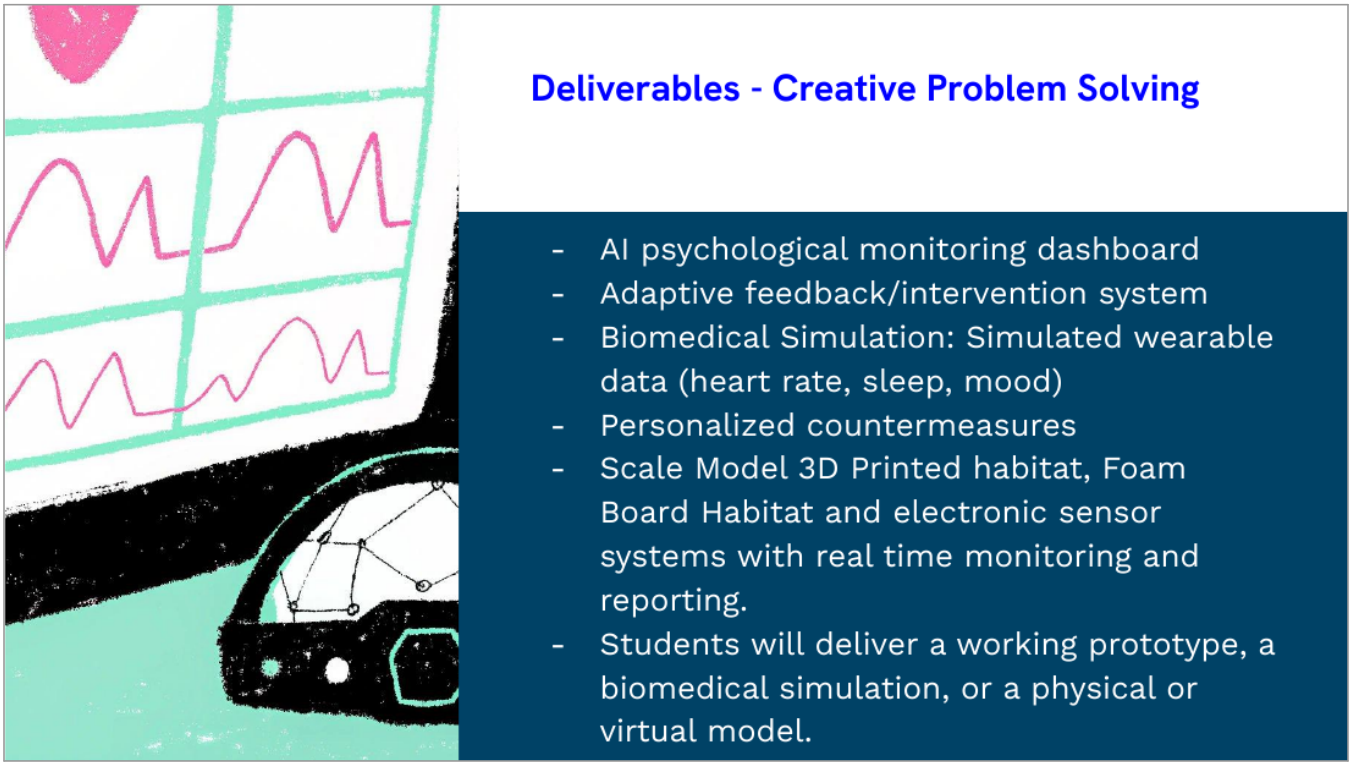
- Long missions = isolation, stress, risk to mental health
  - **Our design:** Every zone and every AI action targets one of the six conditions (anxiety, depression, conflict, fatigue, autonomy, monotony).
- Current habitats focus on physical safety, not psychological needs
  - **Our design:** Our design adds the mental-health layer on top of the safety systems, and safety alarms still beat every comfort setting.



- Monitors crew psychological health with AI
  - **Our design:** The live dashboard (F1) shows all 9 sensors, all 6 scores and the tier for each person.
- Adapts environment to individual needs
  - **Our design:** Changes are made per person and per room: light, sound, temperature, scent, air and schedule.
- Integrates smart building systems (ECLSS, power, environment)
  - **Our design:** These are in the same simulation, with CO2 per zone, power use and alarms.
- Is feasible for lunar deployment
  - **Our design:** The sensors weigh about 13 kg and use about 110 W, and about half the hardware is shared with other systems.



Realistic game: how data gets from a person to the AI



- Working demo of personal space and common spaces
  - **Our design:** The private sleep dome/quarters and the communal galley are both playable in VR.



Realistic game: private sleep dome



Sci-fi game: galley

- AI psychological monitoring dashboard
  - **Our design:** Done (F1 in the realistic game, Menu → Project ARES in the sci-fi game).
- Adaptive feedback / intervention system
  - **Our design:** Done: three tiers, offers you answer yes or no, and crew tasks.
- Biomedical simulation: simulated wearable data (heart rate, sleep, mood)
  - **Our design:** Done: the health check-in shows heart rate, HRV, sleep, skin response and cortisol, and where each reading comes from.



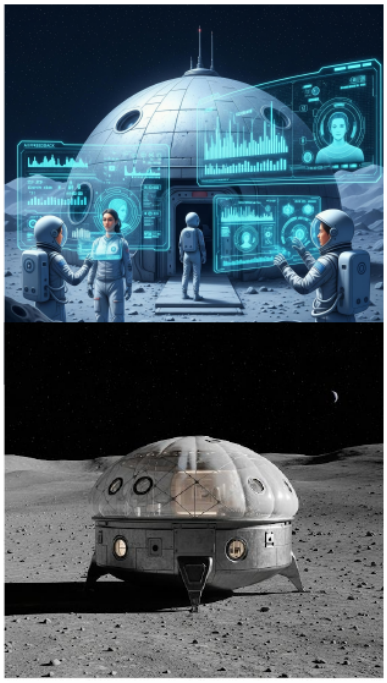
Realistic game: health check-in

- Personalized countermeasures
  - **Our design:** Done: "Help" picks what fits that person's biggest problem, and every morning there's a daily wellness list.



Realistic game: daily wellness list

- Scale model 3D-printed habitat, foam board and electronic sensors
  - **Our design:** The physical model and sensor board are built outside the games (Farhan's model). The games are the virtual model.

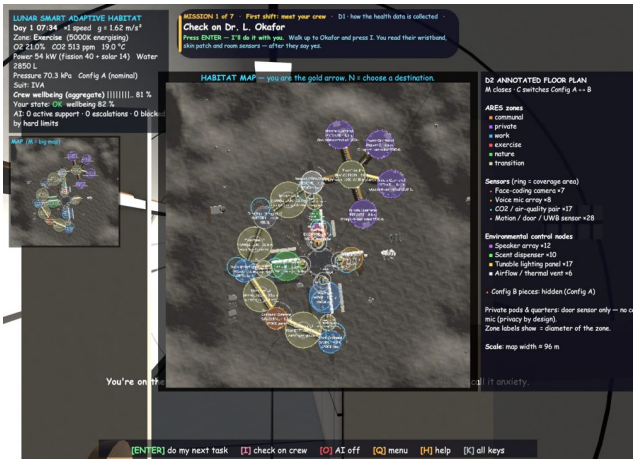


So What would I do First?

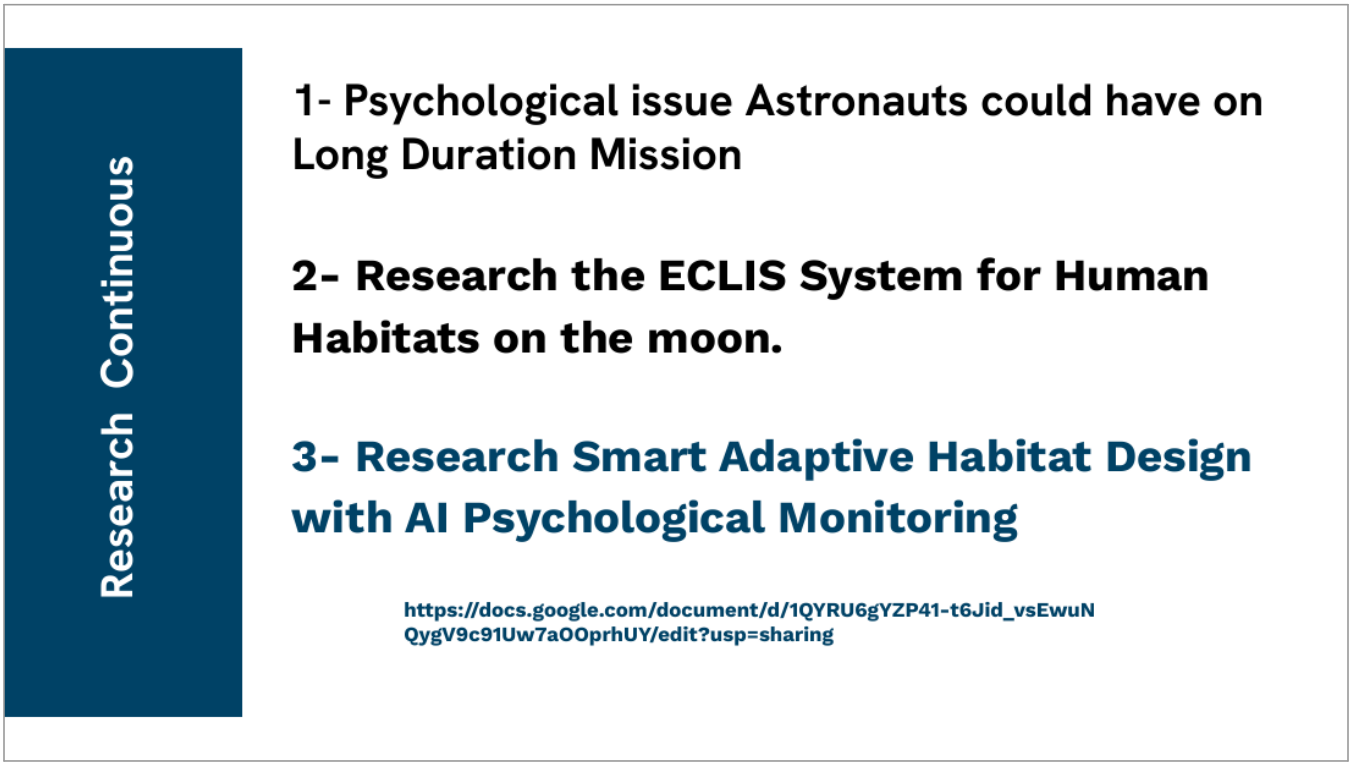
- Research NASA Pill Habitat Design for the moon

Feature	Details (Based on NASA Artemis/Surface Habitat Concepts)
Crew Capacity	2 astronauts (standard), expandable to 4 (peak)
Uncrewed Duration	Up to 6–24 months of autonomous operation
Structure	Multi-level: rigid metallic base + inflatable upper deck
Internal Layout	Sleeping, food prep, hygiene, work, storage
Life Support	Advanced ECLSS, waste management, radiation shielding
Power	Solar arrays + nuclear fission backup
Location	Lunar south pole (e.g., Shackleton Crater)
Autonomy	Remote monitoring, automated maintenance, and mode switching

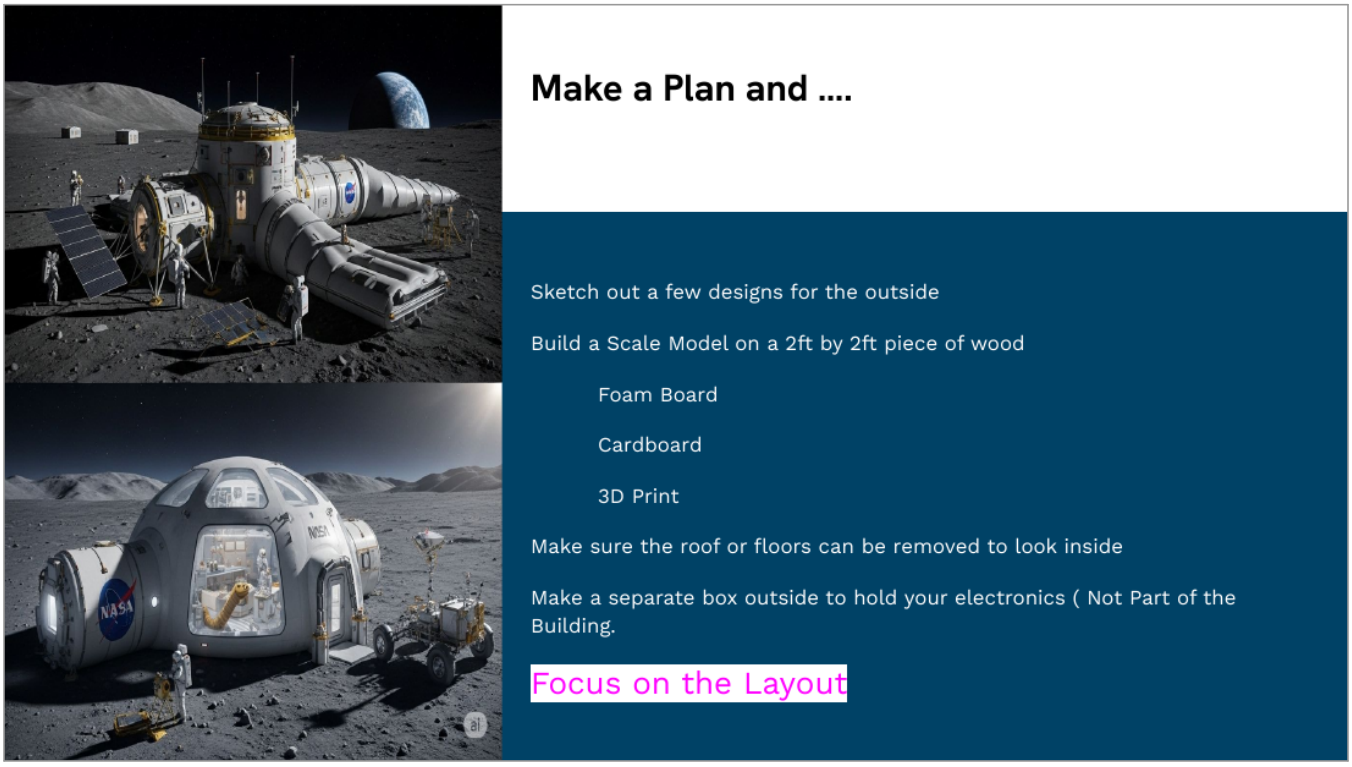
- Crew capacity: 2 standard, expandable to 4
  - **Our design:** We fly 4 (you plus 3), and the design can scale up (see D2).
- Multi-level structure, ECLSS, solar + fission power, south pole
  - **Our design:** Both bases are multi-level, with ECLSS, solar plus a 40 kW fission plant, at the south pole.
- Internal layout: sleeping, food prep, hygiene, work, storage
  - **Our design:** All present, laid out in NASA adjacency order (airlock → hygiene → quarters → galley → work).



Realistic game: map of every zone



- Research continuous
  - **Our design:** Our 64-page research document is where the numbers in the games come from (thresholds, lighting, CO2, room sizes).



- Sketch designs, build a scale model on a 2 ft × 2 ft board
  - **Our design:** The physical model is separate. In Unity you can walk inside every room, and the dimensioned floor plans are in both games.



Realistic game: dimensioned floor plan

- Roof or floors removable to look inside
  - **Our design:** In the game you can use the big map (M), the outside cameras (F10) and third-person view (F) to see inside.





Make a Plan and ....

Sketch out a few designs for the outside

Build a Scale Model on a 2ft by 2ft piece of wood

- Foam Board
- Cardboard
- 3D Print
- Other

Make sure the roof or floors can be removed to look inside

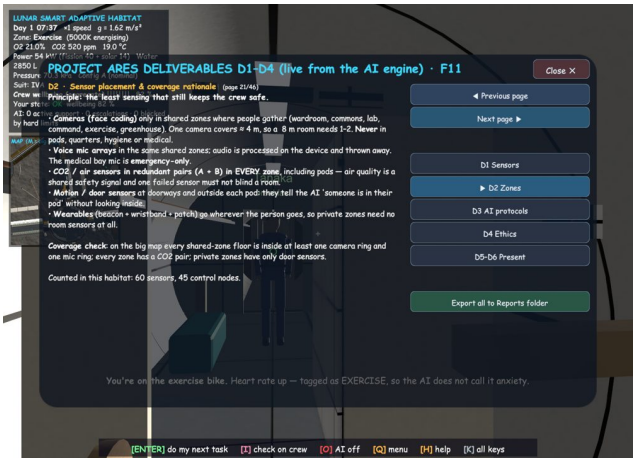
Make a separate box outside to hold your electronics (Not Part of the Building.

Focus on the Layout

- Focus on the layout
  - **Our design:** Rooms follow NASA adjacency, quiet and noisy areas are kept apart, and Config A and B are both available.

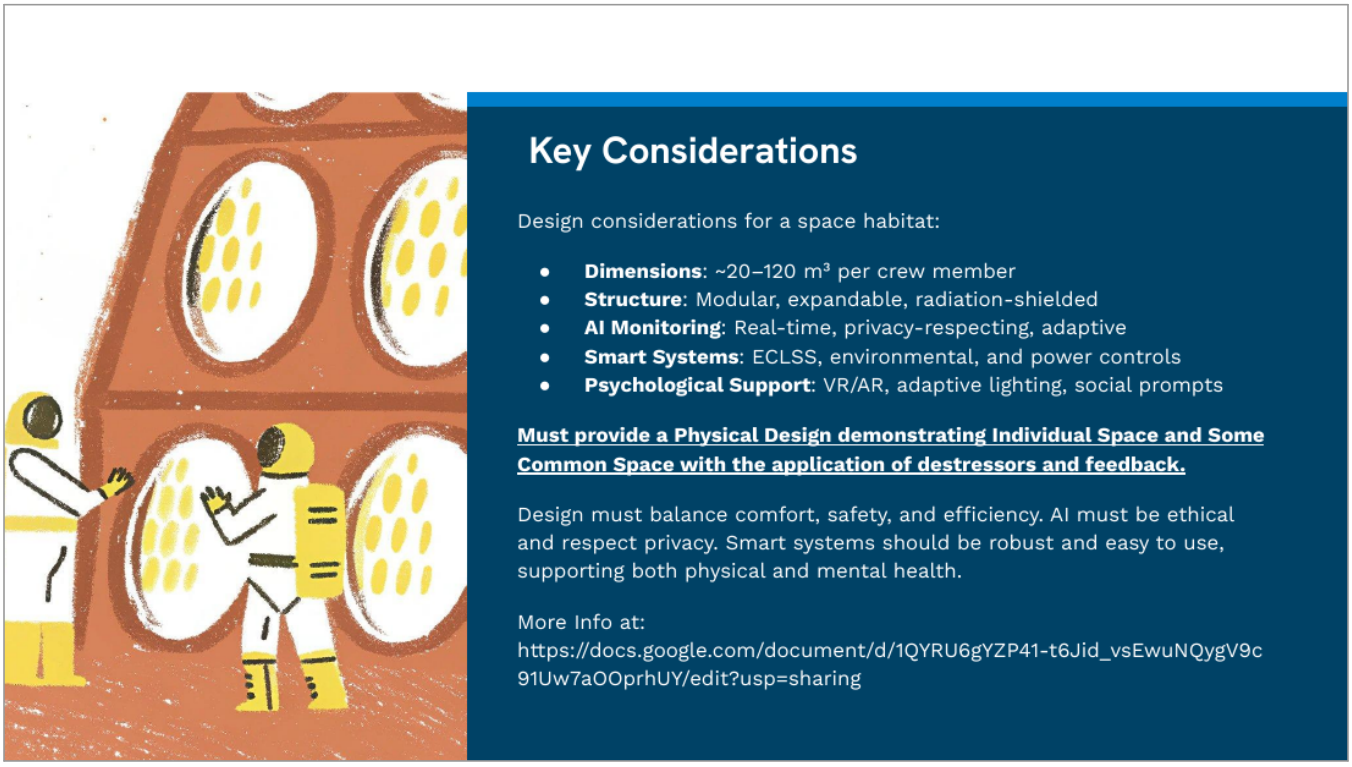


- Smart, adaptive habitat design with AI
  - **Our design:** Both games. The AI adapts each room to the people inside it.
- Psychological monitoring (voice, facial, group interaction, laughter, tears, isolation)
  - **Our design:** Voice and face analysis happen in shared rooms only, processed in the room. Group interaction and isolation come from location beacons (time spent together).

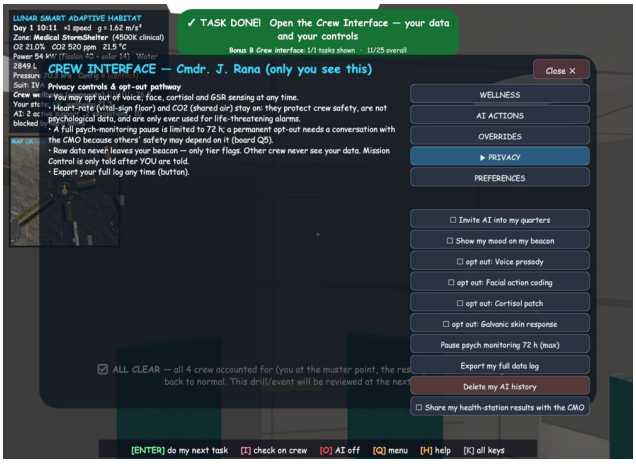


Realistic game: sensor placement

- Focus on one room at a time
  - **Our design:** Each zone has its own design page with its purpose, lighting, sensors and what the AI does there.



- Dimensions ~20–120 m³ per crew member
  - **Our design:** The realistic base gives about 740 m³ per crew member, and about 185 m³ of private space each, which is above the guide.
- Structure: modular, expandable, radiation-shielded
  - **Our design:** Modular nodes and tubes, room for 2 more quarters, and a medical bay that doubles as the storm shelter.
- AI monitoring: real-time, privacy-respecting, adaptive
  - **Our design:** Real-time scores, no sensors in pods, opt-out, export and delete, and hard limits in the code.



Realistic game: privacy controls

- Smart systems: ECLSS, environmental and power controls
  - **Our design:** Done in both games.
- Psychological support: VR/AR, adaptive lighting, social prompts

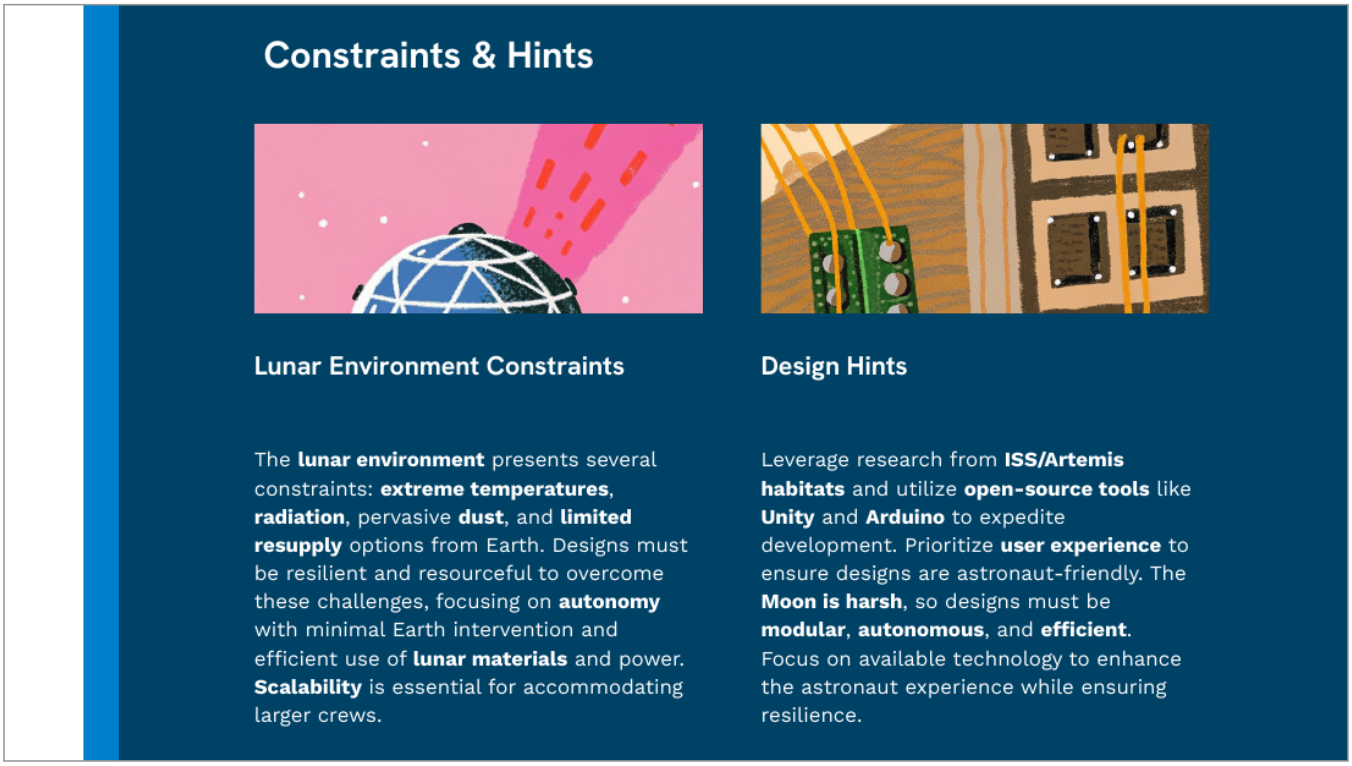
- **Our design:** A VR nature room, circadian lighting, team meals and calls home.

- Physical design showing individual space and common space with destressors
  - **Our design:** The private quarters and pods plus the communal galley, each with its own destressors (light, sound, scent).

- AI must be ethical and respect privacy
  - **Our design:** Our 1,155-word ethics report, 8 hard limits, an AI OFF switch, and the night watch (if anything goes wrong at night, the AI stops and a human takes over).



Realistic game: night watch



- Use Unity and Arduino
  - **Our design:** Built in Unity (both games). The sensor board for the physical model is separate.

- Prioritize user experience, astronaut-friendly
  - **Our design:** One key does the next step (ENTER or G), there's a simple 7-button menu and a short key bar, and both games have a full player manual.



Realistic game: simple menu

- Scalability for larger crews
  - **Our design:** The plan for 6 crew is on the D2 page in both games.



Example Project Flow

Research

Conduct thorough **research** on **psychological factors**, **lunar habitats**, and **smart systems** to lay a solid foundation for the project.

More Info at:
https://docs.google.com/document/d/1QYRU6gYZP41-t6Jid_vsEwuNQygV9c91Uw7aOOprhUY/edit?usp=sharing

Design

Create a detailed **design** by sketching the **layout**, defining **AI parameters**, and planning **integration** to guide the development process.

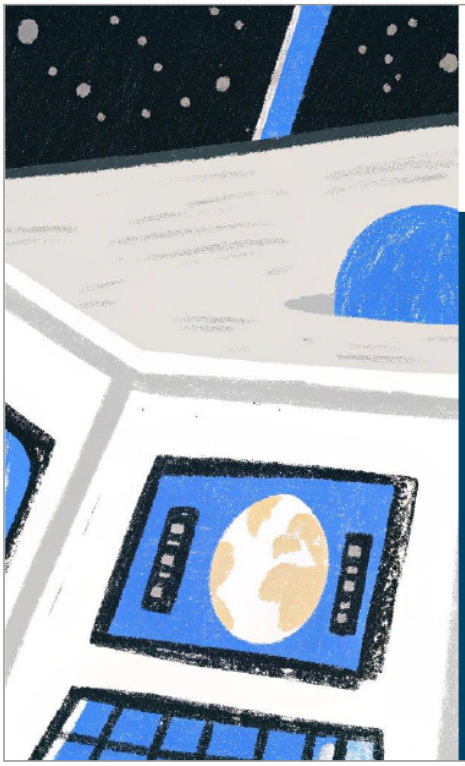
Prototype, Test, Present

Develop a **model/VR** prototype, **develop software**, **simulate scenarios**, **refine interventions**, and then **present** a **demo**, **walkthrough**, and **report**.

- Research
  - **Our design:** Our research document.
- Design
  - **Our design:** D1–D4 design pages inside both games (F11).
- Prototype, test, present
  - **Our design:** Two playable prototypes. We tested every mission and button again and again, ran a 70-mission threshold check and a 72-hour tabletop test, and the presentation run-of-show is built in.



Realistic game: run-of-show



Conclusion

Students innovate for real NASA needs Combining, biomedical, software and engineering skills Supports future lunar explorers' mental and physical health This project is a chance to make a real impact on the future of space exploration. By supporting astronaut well-being, you're helping pave the way for humanity's next giant leap.

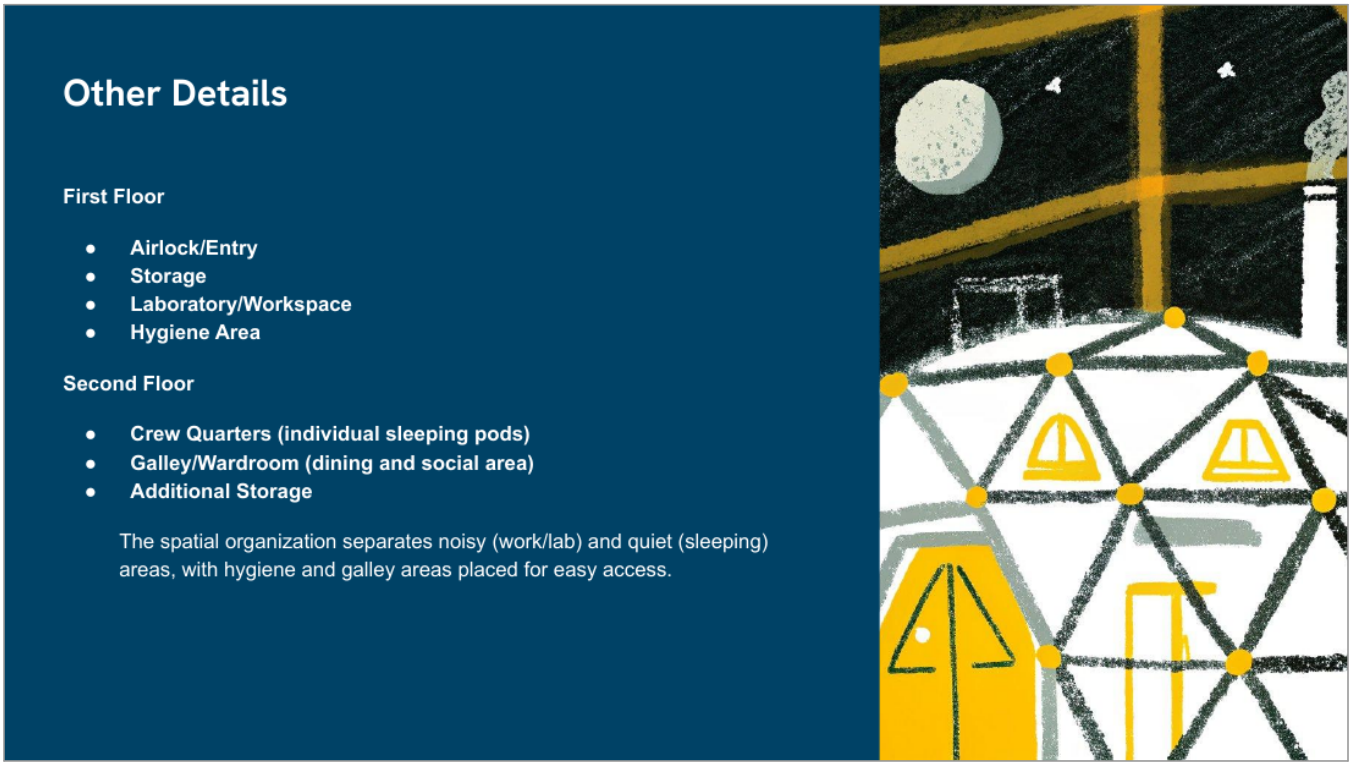
- Conclusion
  - **Our design:** Biomedical, software and engineering come together in one working demo.



- Questions & discussion
  - **Our design:** We practised the six board questions (Q1–Q6) live in the games.



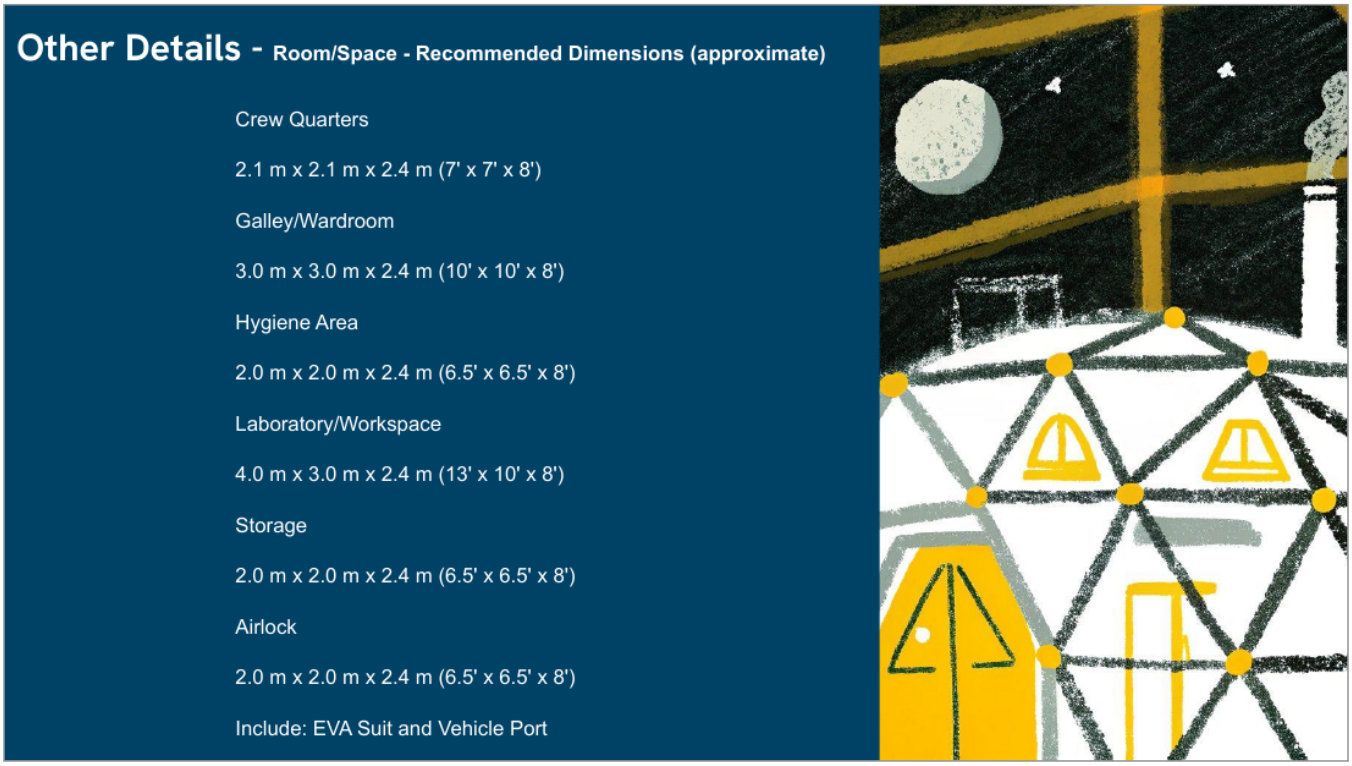
Realistic game: Q2



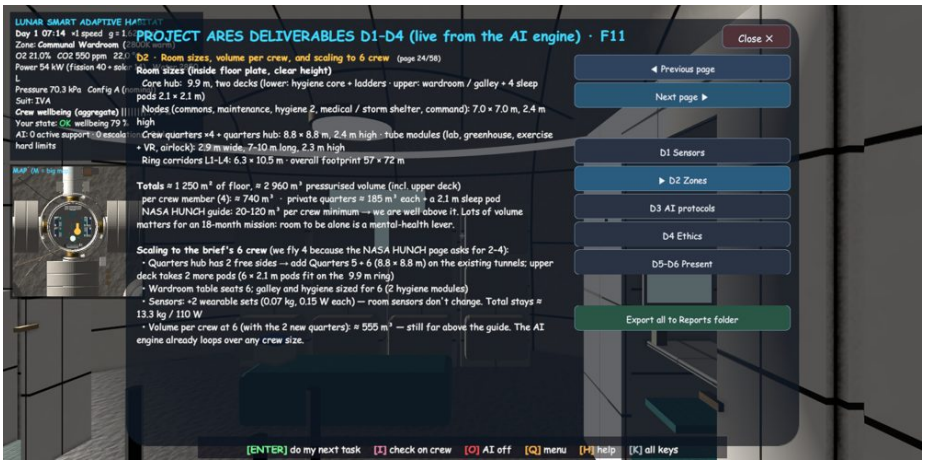
- First floor: airlock, storage, lab/workspace, hygiene
  - **Our design:** In the sci-fi game, level 1 of the pill habitat has the airlock, hygiene, workstations, storage, ECLSS and medical.
- Second floor: crew quarters, galley/wardroom, storage
  - **Our design:** Level 2 has the crew quarters, galley, exercise and sanctuary/VR. In the realistic game, the upper deck of the core hub has the wardroom and 4 sleep pods.
- Separate noisy and quiet areas
  - **Our design:** Sleep areas are away from the lab and work areas, and hygiene and the galley are easy to reach.



Sci-fi game: site plan + pill habitat levels



- Crew quarters 2.1 × 2.1 × 2.4 m
  - **Our design:** Sleep pods are 2.1 × 2.1 m, plus a private room of 8.8 × 8.8 m (2.4 m high) for each person.
- Galley/wardroom 3.0 × 3.0 × 2.4 m
  - **Our design:** The wardroom ring is 9.9 m (bigger, so 4–6 people can eat together).
- Hygiene, storage, airlock 2.0 × 2.0 × 2.4 m
  - **Our design:** Hygiene and storage nodes are 7.0 × 7.0 m, and the airlock module has suitports (EVA suit and vehicle port).
- Laboratory 4.0 × 3.0 × 2.4 m
  - **Our design:** The lab tube is 2.9 m wide and about 10 m long.



Realistic game: room sizes page

Research

Key for this project is to start to Research in:

- **Psychological Support**: VR/AR, adaptive lighting, social prompts
- **AI Monitoring**: Real-time, privacy-respecting, adaptive, Psychological and Physical
- **Smart Systems**: ECLSS, environmental, and power controls
- **Structure**: Modular, expandable, radiation-shielded



- Psychological support: VR/AR, adaptive lighting, social prompts
  - **Our design:** Done (VR nature room, circadian lighting, team meals, calls home).
- AI monitoring: real-time, privacy-respecting, psychological and physical
  - **Our design:** Done (9 sensors, physical vitals always on, mental-health sensors can be turned off).
- Smart systems: ECLSS, environmental and power
  - **Our design:** Done.
- Structure: modular, expandable, radiation-shielded
  - **Our design:** Done (modular nodes, storm shelter, expandable to 6).